

# Point Inside a Convex Polygon

## 1 Problem Definition

### Point-In-Convex-Polygon Test

Given a convex polygon and a query point  $Q$ , decide whether  $Q$  is inside the polygon, outside the polygon, or on its boundary.

Assume the polygon vertices are listed in boundary order:

$$P_1, P_2, \dots, P_n.$$

Each edge is directed from

$$P_i \rightarrow P_{i+1},$$

where the last vertex wraps back to the first:

$$P_{n+1} = P_1.$$

## 2 Same-Sign Orientation Principle

For every directed edge  $P_i P_{i+1}$ , compute

$$s_i = \text{orient}(P_i, P_{i+1}, Q).$$

The value  $s_i$  tells which side of edge  $P_i P_{i+1}$  the query point lies on.

### Decision Rule

|               |                                                       |
|---------------|-------------------------------------------------------|
| all $s_i > 0$ | $\Rightarrow Q$ is inside for counterclockwise order, |
| all $s_i < 0$ | $\Rightarrow Q$ is inside for clockwise order,        |
| mixed signs   | $\Rightarrow Q$ is outside,                           |
| $s_i = 0$     | $\Rightarrow Q$ may be on an edge.                    |

### Core Idea

For a convex polygon, an inside point stays on the same side of every boundary edge.

## 3 Boundary Check

If

$$\text{orient}(P_i, P_{i+1}, Q) = 0,$$

then  $P_i, P_{i+1}, Q$  are collinear. The point is on the segment only if its coordinates lie within the endpoint ranges.

### On-Segment Test

For edge  $AB$ , point  $Q$  lies on segment  $AB$  when

$$\min(a_x, b_x) \leq q_x \leq \max(a_x, b_x)$$

and

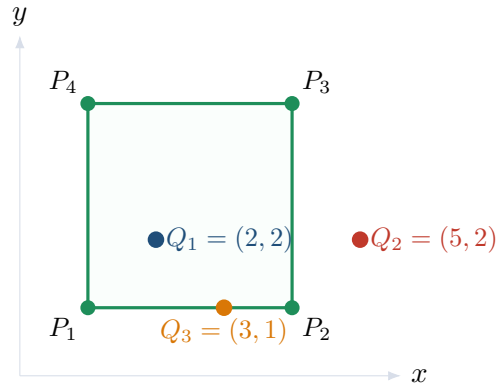
$$\min(a_y, b_y) \leq q_y \leq \max(a_y, b_y).$$

## 4 Example Polygon

Use the square:

$$P_1 = (1, 1), \quad P_2 = (4, 1), \quad P_3 = (4, 4), \quad P_4 = (1, 4).$$

The vertices are ordered counterclockwise.



## 5 Inside-Point Example

### Problem

Test whether

$$Q = (2, 2)$$

is inside the square.

### Solution

For edge  $P_1P_2$ :

$$\overrightarrow{P_1P_2} = (3, 0), \quad \overrightarrow{P_1Q} = (1, 1).$$

$$\text{orient}(P_1, P_2, Q) = 3(1) - 0(1) = 3.$$

For edge  $P_2P_3$ :

$$\overrightarrow{P_2P_3} = (0, 3), \quad \overrightarrow{P_2Q} = (-2, 1).$$

$$\text{orient}(P_2, P_3, Q) = 0(1) - 3(-2) = 6.$$

For edge  $P_3P_4$ :

$$\overrightarrow{P_3P_4} = (-3, 0), \quad \overrightarrow{P_3Q} = (-2, -2). \\ \text{orient}(P_3, P_4, Q) = (-3)(-2) - 0(-2) = 6.$$

For edge  $P_4P_1$ :

$$\overrightarrow{P_4P_1} = (0, -3), \quad \overrightarrow{P_4Q} = (1, -2). \\ \text{orient}(P_4, P_1, Q) = 0(-2) - (-3)(1) = 3.$$

All signs are positive:

$$3, 6, 6, 3.$$

#### Answer

$Q = (2, 2)$  is inside the square.

## 6 Outside-Point Example

#### Problem

Test whether

$$Q = (5, 2)$$

is inside the same square.

For edge  $P_1P_2$ :

$$\overrightarrow{P_1P_2} = (3, 0), \quad \overrightarrow{P_1Q} = (4, 1). \\ \text{orient}(P_1, P_2, Q) = 3(1) - 0(4) = 3.$$

For edge  $P_2P_3$ :

$$\overrightarrow{P_2P_3} = (0, 3), \quad \overrightarrow{P_2Q} = (1, 1). \\ \text{orient}(P_2, P_3, Q) = 0(1) - 3(1) = -3.$$

The signs are already mixed:

$$3, -3.$$

#### Answer

$Q = (5, 2)$  is outside the square.

## 7 Boundary Example

#### Problem

Test whether

$$Q = (3, 1)$$

is inside, outside, or on the boundary of the same square.

For edge  $P_1P_2$ :

$$\overrightarrow{P_1P_2} = (3, 0), \quad \overrightarrow{P_1Q} = (2, 0). \\ \text{orient}(P_1, P_2, Q) = 3(0) - 0(2) = 0.$$

So  $Q$  is collinear with  $P_1P_2$ . Collinearity alone only says that  $Q$  lies on the same infinite line as the edge. To be on the actual edge segment,  $Q$  must also lie between the two endpoints.

Here the edge endpoints are

$$P_1 = (1, 1), \quad P_2 = (4, 1).$$

For a point  $Q = (q_x, q_y)$  to lie on the segment  $P_1P_2$ , its coordinates must satisfy the bounding-box test:

$$\min(1, 4) \leq q_x \leq \max(1, 4), \quad \min(1, 1) \leq q_y \leq \max(1, 1).$$

Since  $Q = (3, 1)$ , this becomes:

$$1 \leq 3 \leq 4, \quad 1 \leq 1 \leq 1.$$

Both conditions are true, so  $Q$  lies between  $P_1$  and  $P_2$ , not merely on the same line.

### Answer

$Q = (3, 1)$  lies on the boundary edge  $P_1P_2$ .

## 8 Pseudocode

### Point in Convex Polygon

```
PointInConvexPolygon(polygon, q):
    positive = false
    negative = false

    for each directed edge (pi, pj):
        value = orientation(pi, pj, q)

        if value > 0:
            positive = true

        else if value < 0:
            negative = true

        else:
            if q lies on segment pi-pj:
                return "boundary"

    if positive and negative:
        return "outside"

    return "inside"
```

## 9 Correctness Intuition

For a convex polygon, every boundary edge acts like a separating line. If  $Q$  is inside, then  $Q$  remains on the same side of every directed boundary edge. If one edge puts  $Q$  on the opposite side, then that edge separates  $Q$  from the polygon interior, so  $Q$  is outside.

### Convexity Requirement

The same-sign orientation test works directly for convex polygons. It should not be applied unchanged to arbitrary non-convex polygons.

## 10 Complexity

The algorithm checks each polygon edge once. For  $n$  polygon vertices, there are  $n$  edges.

time complexity =  $\mathcal{O}(n)$ .

Only a few variables are needed:

auxiliary space =  $\mathcal{O}(1)$ .

## 11 Final Review

### Point-In-Convex-Polygon Summary

$s_i = \text{orient}(P_i, P_{i+1}, Q)$ .

All positive or all negative signs mean inside.

Mixed signs mean outside.

$s_i = 0$  requires an on-segment boundary check.

The method runs in  $\mathcal{O}(n)$  time for a convex polygon.